

1 Energetics: same functional family, different parameterisation

`py_utils/loop_energetics.py` implements the Dudarev (PRL **100**, 135503) prismatic-loop free energy

$$E_l^X(n, T) = P_X(n) \left[\hat{F}_X(T) \ln \frac{4R_X^*(n)}{e\delta} + F_\delta^X + F_c^X \right], \quad \hat{F}_{100}(T) = \hat{F}_{100}^0 \left(1 - \frac{T}{T_c} \right)^{1/4}, \quad T_c = 1185 \text{ K},$$

with $\hat{F}_{111}$ held *temperature-independent* and the single free constant $\hat{F}_{100}^0$ fixed by one calibration target, $\Delta F(n_{\text{ref}}, T^*) = 0$. Gao *et al.* instead compute $\hat{F}$ and F_δ from full anisotropic elasticity with Varshni-fitted C_{11}, C', C_{44} to 1100 K, and condense the result to

$$E_{\text{tot}}(N, T) = \epsilon_0(T) \sqrt{N} \ln N + \epsilon_1(T) \sqrt{N}, \quad \epsilon_i(T) = b_0 + b_1 T + b_2 T^2 + b_3 T^3.$$

Since $P_X \propto \sqrt{N}$ and $\ln R^* = \frac{1}{2} \ln N + \text{const}$, **these are the same functional form**: their Table 3 is a drop-in replacement for your digitised constants. Two structural differences matter:

1. $\frac{1}{2}\langle 111 \rangle$ is held *T-independent* in RadCluster. Gao’s fitted ϵ_0, ϵ_1 for $\frac{1}{2}\langle 111 \rangle$ carry significant linear and cubic T terms. Holding $\hat{F}_{111}$ fixed attributes the *entire* crossover to $\langle 100 \rangle$ softening.
2. RadCluster carries one $\frac{1}{2}\langle 111 \rangle$ variant. Gao resolve $\langle 11\bar{2} \rangle$ and $\langle 1\bar{1}0 \rangle$ side orientations, whose crossover temperatures differ by $\sim 80^\circ\text{C}$ at 100 SIAs — comparable to the effect being modelled.

Numerical comparison. Both treatments agree in *direction*: small loops favour $\langle 100 \rangle$, and the crossover temperature rises with size. The shipped calibration, however, sits high:

	$n = 20$	$n = 50$	$n = 100$	$\Delta F > 0$ at 450°C
RadCluster, shipped ($T^* = 450^\circ\text{C}$ @ $n_{\text{ref}} = 50$)	-13°C	450°C	556°C	$n \leq 49$
Recalibrated $T^* = 455^\circ\text{C}$ @ $n_{\text{ref}} = 100$	—	319°C	455°C	$n \leq 96$
Recalibrated $T^* = 375^\circ\text{C}$ @ $n_{\text{ref}} = 100$	—	215°C	375°C	$n \leq 183$
Gao Fig. 3(b), 100 SIAs	—	—	$375\text{--}455^\circ\text{C}$	—

$\hat{F}_{100}^0 = 0.4799 \text{ eV}/\text{\AA}$ as shipped. Moving the calibration target from $(450^\circ\text{C}, n=50)$ to $(455^\circ\text{C}, n=100)$ costs one line and brings the size-dependence onto Gao’s curve; the $\langle 100 \rangle$ -favoured window at 450°C roughly doubles, $n \leq 49 \rightarrow n \leq 96$. The module docstring already flags the constants as “approximate (digitised from Dudarev Figs. 2–3)” and the size-dependence as a calibration decision — this is the calibration.

2 The kinetic channels: your two, their two

RadCluster carries *three* conversion channels (`py_utils/reaction_rates.py`):

1. $\Gamma_{\text{uni}}(n) = \nu_0 e^{-E_a(n)/k_B T} [1 - e^{-\Delta F/k_B T}]$ — a *direct single-loop rotation*, $E_{a0} = 1.8 \text{ eV}$;
2. the Marian junction $\varphi_{\text{junc}}(n, n') = \varphi_{\text{max}} e^{-\ln^2(n/n')/2\sigma_s^2} \Theta(\min(n, n') \geq n_j^{\text{min}})$ with two-step success $P_{\text{succ}} = e^{-\Delta H_2/k_B T} / (e^{-\Delta H_2/k_B T} + e^{-\Delta H_{\text{rev}}/k_B T})$, $\Delta H_2 = 1.0$, $\Delta H_{\text{rev}} = 0.30 \text{ eV}$;
3. a *size-asymmetric* absorption gate $P_{\text{succ}}^{\text{abs}}$ with its own lower forward barrier ($\Delta H_2^{\text{abs}} = 0.36 \text{ eV}$ in the workbook), justified in the code as the case where an absorbed cluster “inherits — is templated by — the host’s existing $\langle 100 \rangle$ character.”

Gao *et al.* test both loop–loop routes and reach an asymmetric verdict:

- **The symmetric Marian junction is negligible.** “[S]imulation results show that this contribution to the nucleation of $\langle 100 \rangle$ loops is negligible. . . due to the low possibility of two $\frac{1}{2}\langle 111 \rangle$ loops with an identical size and suitable orientations to encounter each other.” Their introduction is blunter: the mechanism “requires the collision between two migrating $\frac{1}{2}\langle 111 \rangle$ loops with similar sizes and overlapping glide cylinders, making it unlikely in reality.”

- **The asymmetric reaction dominates.** Their conclusion (5) identifies “ $\frac{1}{2}\langle 111 \rangle + \langle 100 \rangle$ ” as “the origin causing the SIA transfer between loops of different types,” with the resultant character following the *larger* loop (after Arakawa’s in-situ TEM).

Your channels (2) and (3) map exactly onto their negligible and dominant routes. So your reading that both are needed is consistent with their physics — **but the weighting is suspect:** φ_{junc} is Gaussian in $\ln(n/n')$, *peaked at* $n = n'$, and additionally gated by $\min(n, n') \geq n_j^{\text{min}}$. It maximises precisely the encounter Gao find improbable. **One-line test:** set `phi_max_junc` = 0 and check whether the $\langle 100 \rangle$ fraction survives on the absorption channel alone. If it does, the junction is a tuning knob, not a mechanism; if it collapses, you have a defensible disagreement with Gao worth stating explicitly.

3 The unary channel is the one experiment does not support

Gao are explicit: “The population re-arrangement proceeds *not* because a single $\frac{1}{2}\langle 111 \rangle$ loop directly transfers its orientation into the $\langle 100 \rangle$ configuration. Instead, loop mutual-interactions provoke the population modification.” Yao’s heavy-ion experiments saw no direct transformation — only the abrupt disappearance of $\frac{1}{2}\langle 111 \rangle$ above 465°C.

Γ_{uni} is a direct single-loop rotation. Your own code comment concedes it is calibrated rather than derived (“calibrated to place the $f_{111}(T)$ crossover at $\sim 350^\circ\text{C}$ at reactor dose rates”). That is defensible as a coarse-grained effective channel standing in for unresolved processes — but it should be labelled as such, because a referee holding these two papers will read it as the mechanism that experiment rules out.

4 What Gao have that RadCluster does not

- **C15 Laves clusters — the single largest gap.** In Gao’s model $\langle 100 \rangle$ loops “nucleate mainly via the collapse of C15 clusters”; C15 also acts as an SIA buffer that “significantly reduces the density of $\langle 100 \rangle$ loops” and modifies loop reaction rates. RadCluster has no C15 population at all, so every $\langle 100 \rangle$ must come from conversion. That is a genuinely different nucleation hypothesis, not a missing detail.
- **Cascade overlap with pre-existing clusters** (their F_{ol} terms, Eq. 9) as a transformation channel among $\frac{1}{2}\langle 111 \rangle$, $\langle 100 \rangle$ and C15.
- **Annular reaction cross-sections.** They state the spherical reaction volume “became invalid for the 2-dimensional loops with sizes over a few nanometers” and adopt annular cross-sections to capture coaxial-loop strain repulsion. RadCluster uses $A_{\text{sph}}m^{1/3}$ and $A_{\text{loop}}n^{1/2}$ capture-radius kernels throughout.
- **Trap-mediated loop diffusion** (Kohnert & Wirth), $D_l^{\text{dt}} = \nu_l^{\text{dt}}\lambda_l^2/2N$ with $\nu_l^{\text{dt}} \propto$ dose rate — an irradiation-driven random walk. RadCluster uses a thermal $D_n^{\text{1D}} \propto n^{-s}e^{-E_m/k_B T}$ with a rotational-correlation factor.
- **Cascade cluster spectrum:** exponential, $f^{\text{SIA}}(n) = 0.89e^{-0.67n}$ ($\sim 54\%$ clustered), against your power law $C_i n^{-s_i}$, $f_i^{\text{cl}} = 0.58$. Similar clustered fraction, very different large-cluster tail.

Conversely, Gao exclude vacancy-type loops and report that voids barely form; RadCluster’s coupled cavity-loop system covers a regime their model does not.

5 The strongest point of agreement

Your experimental database encodes a *dose*-driven transition (`loop_burgers_fraction.py`: “a/ $\frac{1}{2}\langle 111 \rangle$ dom. (<4 dpa); a/ $\langle 100 \rangle$ dom. (>4 dpa)”), and Gao’s Fig. 14 shows the same — the $\frac{1}{2}\langle 111 \rangle$ SIA fraction falling steeply with dose at fixed temperature, reaching ~ 0 by 0.5 dpa at 500°C. Both therefore say the transition is not purely thermodynamic. Gao’s conclusion (7) puts it as temperature acting *through diffusivity*: “most $\frac{1}{2}\langle 111 \rangle$ loops are merged with $\langle 100 \rangle$ loops during their long-range movements when the temperature is over (equal) 500°C.” That is an argument for weighting your absorption channel over

both the unary and junction channels — and the dose dependence is the observable that discriminates them, since a thermodynamic crossover alone predicts none.

Recommended actions. (i) Recalibrate `LoopEnergetics` to $(455^\circ\text{C}, n_{\text{ref}} = 100)$ or import Gao’s Table 3 directly, and give $\frac{1}{2}\langle 111 \rangle$ its T -dependence. (ii) Run the $\varphi_{\text{max}} = 0$ sensitivity to establish whether the junction is load-bearing. (iii) Relabel Γ_{uni} as an effective channel, or drop it and test whether absorption alone reproduces $f_{111}(T)$. (iv) Decide explicitly whether the absence of C15 is a modelling choice you defend or a gap you acknowledge — it is the first thing these authors will look for.